25900562 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Elective (DSE) - MG3DSEPHY204 - INTRODUCTION TO MATERIALS SCIENCE

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Type Questions

Answer any **10** questions

Each question carries **2** marks

1. What do you mean by smart materials? Which are the two components of smart materials? [U] / [CO1]
2. Why are amorphous solids considered isotropic in nature? [U] / [CO1]
3. Find the Miller indices of a plane that intercepts the axes at (a, a, a). [U] / [CO1]
4. Schematically plot attractive, repulsive and net force versus interatomic separation for two atoms or ions. Also mark the equilibrium distance [U] / [CO2]
5. What is solvus line? [K] / [CO4]
6. What is dendritic structure? [K] / [CO4]
7. Differentiate between homogenous and hetrerogenous nucleation [U] / [CO5]
8. What is the effect of higher activation energy on transformation kinetics? [U] / [CO5]
9. What is the critical radius of a nucleus, and what does it signify? [U] / [CO5]

10. What role does undercooling or supercooling play in controlling phase transformation kinetics? [U] / [CO5]

11. Name the test used to measure notch toughness. [K] / [CO6]

12. Explain Hooke's law? [K] / [CO6]

13. Define brittle fracture. [K] / [CO6]

14. Define corrosion and give one common example. [K] / [CO6]

[2x10 = 20]

Part B

Short Essay Type Questions

Answer any **6** questions

Each question carries **5** marks

15. Compare ionic and covalent bonding with respect to bond energy, melting point, and conductivity. [U] / [CO1]

16. Explain the seven crystal systems with their axial relationships [U] / [CO1]

17. Discuss how impurities influence diffusion processes in solids. Support your answer with relevant examples. [U] / [CO3]

18. What is the difference between the states of phase equilibrium and metastability? [U] / [CO4]

19. Apply the Gibbs Phase Rule to a binary phase diagram and discuss how it helps in determining the number of degrees of freedom in different regions of the diagram. [U] / [CO4]

20. Explain how the compressive residual stresses are beneficial for fatigue resistance. [U] / [CO6]

21. With examples, give the applications of materials with high ductility and low ductility. [U] / [CO6]

22. Explain the mechanism of oxidation in metals and derive the parabolic rate law for oxidation kinetics. Discuss its significance in understanding corrosion resistance. [U] / [CO6]

[5x6 = 30]